

Porter Delivery Data Analysis & Dashboard Project Report

Project Overview

This project analyzes Porter delivery operations using Microsoft Excel. The objective was to clean the raw dataset, perform exploratory data analysis (EDA), derive business insights, and build an interactive dashboard for operational decision-making.

Objectives

- Analyze order distribution across markets.
- Measure delivery performance across store categories.
- Identify peak ordering hours and weekly demand trends.
- Evaluate operational efficiency using statistical analysis and correlations.
- Build an interactive Excel dashboard using Pivot Tables, Pivot Charts and Slicers.

Dataset Summary

- Total Orders: 197,386
- Markets Covered: 6
- Delivery Time Metric: delivery_duration_min
- Data Cleaning Included: Missing values, invalid values, delivery duration calculation, outlier removal (>300 minutes), helper columns and derived metrics.

Tools & Techniques

Microsoft Excel, Pivot Tables, Pivot Charts, Conditional Formatting, Slicers, Correlation Analysis (CORREL), Standard Deviation, Helper Columns, KPI Cards and Dashboard Design.

Key Business Insights

- Market 2 generated the highest order volume.
- Average delivery time after cleaning: 47.62 minutes.
- Only 13.14% of deliveries were completed within 30 minutes.
- Peak order demand occurred during early morning hours, with a secondary evening surge.
- Delivery performance varies across store categories more than across markets.
- Price range and subtotal showed a moderate positive correlation (~0.51), indicating higher price diversity often leads to higher order values.
- Partner utilization showed only a very weak relationship with delivery time, suggesting other operational factors influence delivery performance.

Dashboard Features

- Five KPI Cards
- Four Interactive Charts
- Three Dynamic Slicers (Market, Store Category, Order Protocol)

- Fully interactive PivotChart-based dashboard

Project Limitations

The dataset did not contain a unique Customer ID. Therefore, first-time versus repeat customer analysis could not be performed without making unsupported assumptions. This limitation has been documented rather than fabricating results.

Conclusion

The project demonstrates an end-to-end Excel analytics workflow including data cleaning, statistical analysis, business interpretation, KPI development, and dashboard creation. The dashboard provides decision-makers with actionable insights into order demand, delivery performance, and operational efficiency while maintaining a professional, portfolio-ready presentation.